MOHAMMED ISLAM

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Kubernetes & AWS Specialist — Senior Platform Engineer

SUMMARY

Senior Platform Engineer with 7+ years provisioning and maintaining production Kubernetes clusters across AWS, GCP, and Azure. Expert in designing enterprise-scale containerized environments, implementing IaC with Terraform, and leading complex production issue resolution. Proven track record providing technical leadership, coaching junior engineers, and eliminating toil through automation. AWS-certified at Professional Specialty level with extensive Agile framework experience and active SC clearance.

CERTIFICATIONS

- **AWS Certified Fellow** - All 10 AWS Certifications at Professional & Specialty Level
- **AWS Certified Professional Architect**
- **AWS Certified Speciality Security Engineer**
- **AWS Certified Speciality Network Engineer**
- **AWS Certified Speciality Machine Learning Engineer**
- **HashiCorp Certified: Terraform Associate**

PROJECT HIGHLIGHTS

- **Kubernetes Expertise:** Extensive experience with EKS, and Google Kubernetes Engine orchestration platforms, specializing in production cluster design and management with Cluster API
- **Enterprise Migrations:** Led end-to-end PagerDuty to OpsGenie migration coordinating 20+ stakeholder teams, delivering $150K annual savings with 100% feature parity
- **Container Orchestration:** Advanced proficiency in Docker containerization and Kubernetes deployment strategies including canary and rolling deployments with KEDA autoscaling
- **Cloud-Native Development:** Proficient in Python and Golang for developing Kubernetes operators, controllers, and automation tools with comprehensive observability integration
- **Infrastructure Migrations:** Successfully migrated 50+ applications from VM-based infrastructure to containerized Kubernetes environments using Cluster API

CORE KUBERNETES & CONTAINER CAPABILITIES

Kubernetes Platforms & Distributions

- Amazon Elastic Kubernetes Service (EKS)
- Google Kubernetes Engine (GKE)
- Red Hat OpenShift Container Platform
- On-premises Kubernetes clusters on RHEL infrastructure

Container & Orchestration Technologies

- Docker containerization and multi-stage builds on RHEL environments
- Kubernetes API development and custom resources
- Helm chart development and management
- Kustomize for configuration management
- Istio service mesh implementation

Unix/Linux System Administration

- Shell scripting with Bash for automation and monitoring
- Load balancer configuration including Nginx Ingress Controller & HAProxy implementations
- Infrastructure as Code (IaC) principles with Terraform and Ansible

Kubernetes CI/CD & Automation

- GitOps workflows with ArgoCD
- Jenkins pipeline orchestration for container builds and deployments
- GitHub Actions and Git-based CI/CD pipelines

OBSERVABILITY & MONITORING EXPERTISE

Outstanding Kubernetes and observability background with **Prometheus, Mimir, Loki** and **Grafana, APM** and tracing with **Otel and NewRelic**, log pipelines with **Fluentd/bit, Vector** on **ELK stack**. Extensive experience with **OpenSearch, Elasticsearch** and time-series databases. Service mesh experience with **Istio** for secrets management. Load balancer configuration with **HAProxy** and ingress controllers **Nginx Ingress Controller** with monitoring expertise around **Otel Collector**. Built comprehensive CI/CD solutions using **ArgoCD, GitLab, GHA, Jenkins and Ansible Automation Platform**.

PROFESSIONAL EXPERIENCE

Phillip Morris International — June 2024 - Present

Senior Platform Engineer - Kubernetes & Cloud Infrastructure

- Architected enterprise-scale Kubernetes platform using Cluster API with integrated day-2 operations tooling, providing 50+ development teams with standardized, secure cloud infrastructure
- Led end-to-end PagerDuty to OpsGenie migration in 6 months, ensuring 100% feature parity while coordinating 20+ stakeholder teams and delivering $150K in annual cost savings
- Developed Node.js middleware application integrating ServiceNow incident data with real-time market-location-based status updates, enabling executive leadership monitoring across global operations
- Designed declarative infrastructure provisioning platform using Terraform modules and custom Golang tooling, enabling complete cloud environment deployment through single YAML configurations
- Implemented OIDC federated authentication for service repositories with External Secrets Operator integration, eliminating manual secret management across 50+ microservices
- Engineered automated disaster recovery testing framework using Point-in-Time Recovery (PITR), reducing manual testing effort by 80% and ensuring 4-hour recovery objectives

HMRC — January 2023 - March 2024

Site Reliability Engineer - Platform Squad (Kubernetes Specialist)

- Architected and delivered platform using Cluster API with integrated day-2 operations tooling on Linux infrastructure, providing 50+ development teams with standardized, pen-tested secure cloud infrastructure
- Successfully migrated 50+ applications from VM-based infrastructure to containerized Kubernetes environments, reducing deployment time from weeks to hours
- Implemented standardized compliance testing through Open Policy Agent and Conftest with Bash automation scripts, consolidating CI/CD runners achieving 70% cost reduction
- Built PCI-compliant Ansible playbook for automated security hardening of Amazon CIS Benchmark AMIs on RHEL systems, ensuring consistent security posture across all EC2 instances
- Architected automated SSH key management system using AWS Lambda with 60-day rotation cycles, improving security compliance across 200+ EC2 instances
- Designed DynamoDB-based AMI status tracking system for multi-account environments, providing real-time visibility and reducing troubleshooting time by 50%

- Implemented automated Cluster API node evacuation system for availability zone maintenance, enabling zero-downtime disaster recovery testing

William Hill June 2022 - January 2023
Site Reliability Engineer - Kubernetes Platform

- Architected self-service Prometheus deployment framework with Thanos sidecar integration, enabling 30+ development teams to implement standardized monitoring within Cluster API managed clusters
- Built custom alerting framework using Prometheus kube-state-metrics and team-based labeling with Bash scripting, enabling automated incident routing and reducing alert response time by 60%
- Optimized OpenSearch cluster performance through JVM heap tuning and garbage collection optimization on RHEL systems, improving query response times by 35%
- Developed custom Prometheus exporters for AWS services integration using Golang, consolidating monitoring data into unified dashboards for improved service visibility

Nationwide Building Society September 2017 - June 2022
Site Reliability Engineer - Enterprise Kubernetes Team *2020 - June 2022*

- Implemented Gatekeeper policy enforcement with Python Behave testing framework, preventing unauthorized root access and achieving 100% penetration test compliance
- Deployed Karpenter with AWS Spot Instances to replace cluster-autoscaler in Cluster API managed environments, reducing compute costs by 60%
- Built AWS Glue ETL pipeline to centralize Kubernetes cluster performance metrics, reducing performance regression incidents by 40%
- Implemented automated lab cluster lifecycle management using Cluster API with evening shutdowns, reducing non-production infrastructure costs by 85%
- Automated CoreDNS performance validation post-cluster deployment, ensuring 99.9% DNS resolution SLA compliance across customer microservice deployments

Platform Engineer - Observability & Monitoring Team *2018 - 2020*

- Deployed Grafana Labs observability stack (Loki, Mimir, Tempo) with comprehensive k6 load testing, ensuring 99.9% availability while reducing infrastructure costs by 30%
- Implemented KEDA autoscaling for Kubernetes components with optimized Loki query processing using Golang microservices, reducing monitoring infrastructure costs by 40%
- Built self-service Grafana dashboard provisioning pipeline using Helm templating and automated CI/CD promotion, reducing support requests by 90%

Platform Engineer - Digital Transformation *September 2017 - 2018*

- Led 12-person engineering team to build containerized MERN stack application using Infrastructure as Code for enterprise digital transformation initiative

EDUCATION

University of Warwick September 2013 - August 2017
Department of Engineering - MEng Systems Engineering

Final Year Project

- *Neural Network Modeling of Battery Degradation* - Developed machine learning models to predict lithium-ion battery capacity fade using supervised learning algorithms and time-series analysis, achieving 92% prediction accuracy for capacity retention over 1000+ charge cycles.